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Key Idea:

- For each bit position b:
- Let count1 = how many numbers have bit b=1
  count0 = N - count1
- Now, any pair (i,j) differs at bit b if one has 1 and other has 0.

Example: A = [1, 3, 5]
Binary (3bits)

- 1 = 001
- 3 = 011
- 5 = 101

Bit 0 : values = 1, 1, 1
count1 = 3
count0 = 0
contribution = 2X 3X 0 = 0

Bit 1 : values = 010
count1 = 1
count0 = 2
contri = 2X1X2 = 4

Bit 2 : values = 001
count1 = 1
count0 = 2
contri = 2X1X2 = 4

Result = 0+4+4 = 8

Example 2: A = [2,3]

- 2 = 10
- 3 = 11

Bit 0 = value = 01

count1 = 1

count0 = 1

contri = 2 × 1 × 1 = 2

Bit 1 = value = 11

count1 = 2

count0 = 0

contri = 2 × 2 × 0 = 0

Result = 2 + 0 = 2

ans

[1,2,3,

9 = 512

2^{10} = 1024

100 = 225